

Real-time measurement of idiosyncratic risk for public US firms

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To what extent does corporate idiosyncratic risk evolve over time? As undiversifiable systematic risk influences corporate valuations, a firm's idiosyncratic shocks also contribute to its valuations. Understanding these firm-specific risks is crucial for creating trading strategies for investors. This short paper describes a composite index for measuring the combined idiosyncratic risk from the top seven U.S. public firms, namely Amazon, Apple, Google, Meta, Microsoft, Nvidia, and Tesla.

The econometric framework for constructing the composite index using both corporate finances and macro indicators applies a dynamic factor model. The dataset includes corporate balance sheets, stock prices, and macroeconomic fundamentals. These series, except monthly series of stock prices, interest rates, and inflation, are quarterly. All series run from January 2015 until April 2026. The macroeconomic indicators used are log real GDP per capita, the inflation rate expressed in percent per year, and the nominal interest rate expressed in percent per year and derived from 3-month Treasury bills. All series are downloaded from FRED. The stock prices sourced from Yahoo Finance are aggregated to a monthly level and are then expressed in logs.

To provide evidence of idiosyncratic risk, I first construct common factors affecting these top 7 firms using principal component analysis (PCA). This technique allows for multiple common macroeconomic factors affecting all of them. I extract five principal components of log stock

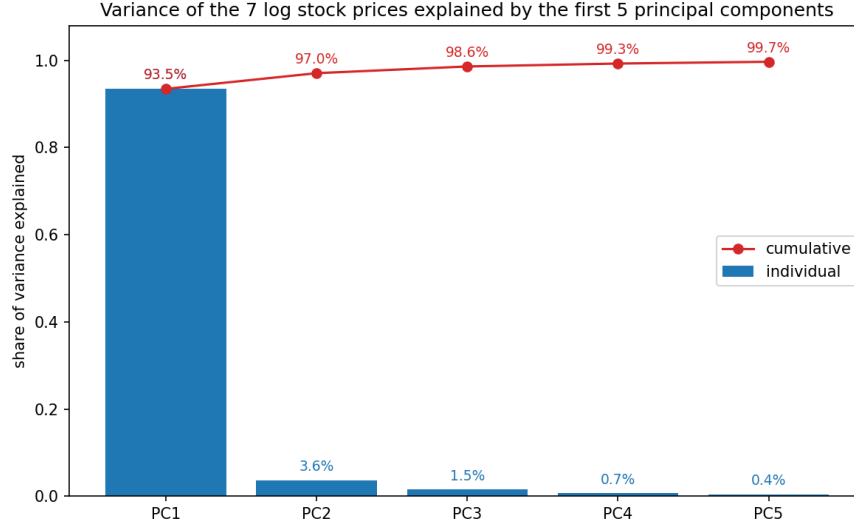


Figure 1: Percentage of variance accounted for by each factor

prices as below.

$$p_t = \sum_{i=1}^5 \lambda_i f_{i,t} + \nu_t \quad (1)$$

where p_t is a 7×1 vector of monthly firm-level equity prices at time t between January 2015 and April 2026. λ_i is the 7×1 vector of firm-level factor loadings on the i -th common factor. ν_t is the vector of residuals. I find that systematic risk from the first principal component explains over 90 percent of the variation in stock prices. As shown in Figure 1, the first five factors explain nearly all of the variation.

I compile a panel of corporate financial variables from public SEC Financial Statement Data Sets¹. The SEC extracts common fields from public filings, including 10-K and 10-Q filings. I construct eight standard variables commonly used in financial analysis. They are revenue, cost of revenue, gross profit, operating income, R&D expense, net income, total assets, and stockholders' equity.² Table 1 describes these variables in detail. Unlike the common practice of directly reporting R&D, Amazon reports Technology and Infrastructure. I omit the R&D expense for Amazon. To visualize how these financial variables explain idiosyncratic risk, I construct a PCA with only one factor and extract the residual. Then I estimate the correlation between each financial variable

¹See <https://www.sec.gov/data-research/sec-markets-data/financial-statement-data-sets>

²Operating income and net income enter in levels, and other variables are log transformed.

Table 1: Quarterly firm financial variables.

Variable	Description
Revenue	Total net sales (top line)
Cost of revenue	Direct cost of goods and services sold
Gross profit	Revenue less cost of revenue
Operating income	Profit from operations before interest and taxes
R&D expense	Research and development spending
Net income	Bottom-line profit after interest and taxes
Total assets	Book value of all assets (period end)
Stockholders' equity	Book value of shareholders' equity (period end)

and the residual term for each firm. Figure 2 indicates the explanatory power between corporate financials and the residual of the single-factor model. This indicates that these financial variables potentially explain firm idiosyncratic risk. This motivates me to include these financial variables in a dynamic factor model.

Let P_{jt} be a j -dimension vector, where j indexes different series: stock prices (7 series), macroeconomic indicators (3 series), and financial measures ($8 \times 6 + 7$ series), which exclude R&D expense for Amazon. Index the M observable series by $j = 1, \dots, M$.

Let c index the firm to which the stock price and financial measure belong. Every series loads on a common national factor F_t^N . Additionally, firm-level series load on a firm-specific factor F_t^c . The indicator function $\mathbb{I}_c$ ensures that only the factor specific to firm c influences that firm.

$$P_{jt} = \lambda_j^N F_t^N + \sum_{c=1}^7 \lambda_j^c F_t^c \mathbb{I}_c + u_{jt}, \quad \text{firm-level series} \quad (2)$$

$$P_{jt} = \lambda_j^N F_t^N + u_{jt}, \quad \text{macro indicators} \quad (3)$$

Moreover, I assume that the error term follows an AR(1) process, while all factors follow AR(2)

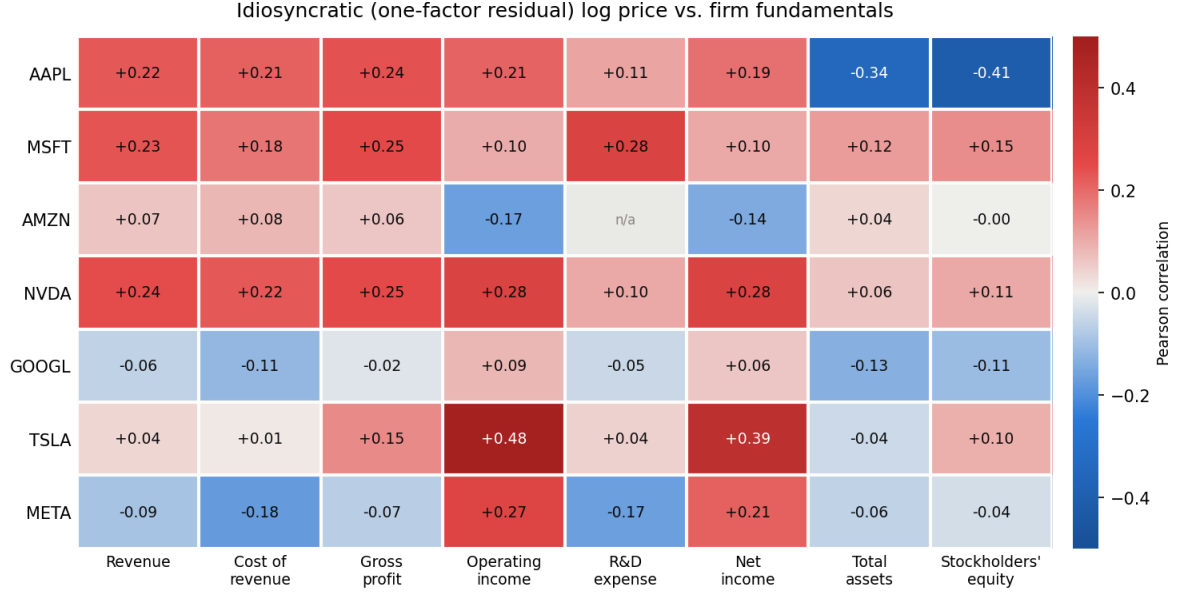


Figure 2: Correlation between each firm's idiosyncratic one-factor residual log stock price and its eight quarterly financial measures.

processes.

$$F_t^b = \phi_1^b F_{t-1}^b + \phi_2^b F_{t-2}^b + \eta_t^b, \quad b \in \{N, 1, \dots, 7\} \quad (4)$$

$$u_{jt} = \rho_j u_{j,t-1} + \varepsilon_{jt}, \quad \varepsilon_{jt} \sim \mathcal{N}(0, \sigma_j^2). \quad (5)$$

I cast the system on a monthly basis. Whenever quarterly variables are not observed, I follow Mariano and Murasawa (2003)³ by treating missing monthly data as loading on the same latent monthly factors.

I use z-score standardization for all series mentioned above and stack them into a state-space form. I use `DynamicFactorMQ` in the `statsmodels` package to estimate the model. The focus below is on the firm-specific factors $\{\hat{F}_t^c\}_{c=1}^7$.

To make the latent factors easy to interpret, I first orient every factor to be *negatively* correlated with real output per capita. The composite index rises when output falls. Each oriented factor is further standardized to zero mean and unit variance, resulting in final factors $\tilde{g}_{ct}$.

³Mariano, R. S., & Murasawa, Y. (2003). A new coincident index of business cycles based on monthly and quarterly series. *Journal of Applied Econometrics* (Chichester, England), 18(4), 427–443.

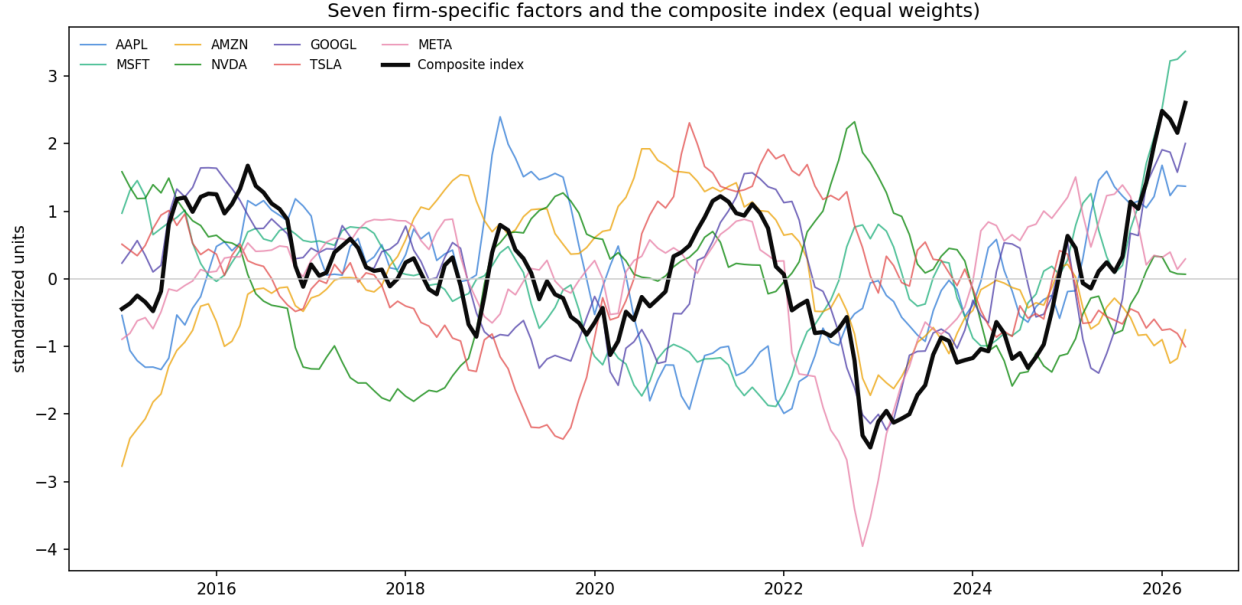


Figure 3: Seven standardized firm-specific factors and the equal-weight composite index (January 2015–April 2026)

I construct the index under three different weighting methods.

$$\text{Equal weights: } C_t^{\text{EW}} = \frac{1}{7} \sum_{c=1}^7 \tilde{g}_{ct}, \quad (6)$$

$$\text{PCA weights: } C_t^{\text{PCA}} = \sum_{c=1}^7 w_c \tilde{g}_{ct}, \quad w_c = \frac{\ell_{1c}^2}{\sum_j \ell_{1j}^2}, \quad (7)$$

$$\text{Geometric: } C_t^{\text{GM}} = \exp\left(\frac{1}{7} \sum_{c=1}^7 \ln m_{ct}\right), \quad m_{ct} = 1 + 99 \frac{\tilde{g}_{ct} - \min_t \tilde{g}_{ct}}{\max_t \tilde{g}_{ct} - \min_t \tilde{g}_{ct}}. \quad (8)$$

where ℓ_1 is the vector of loadings of the first principal component of the seven normalized factors.

Figure 3 plots the equal-weighted composite index against the seven underlying firm-specific factors. The index traces the common tendency of the firm factors while smoothing firm-specific cycles.

Figure 4 plots all three indices from different weighting methods.

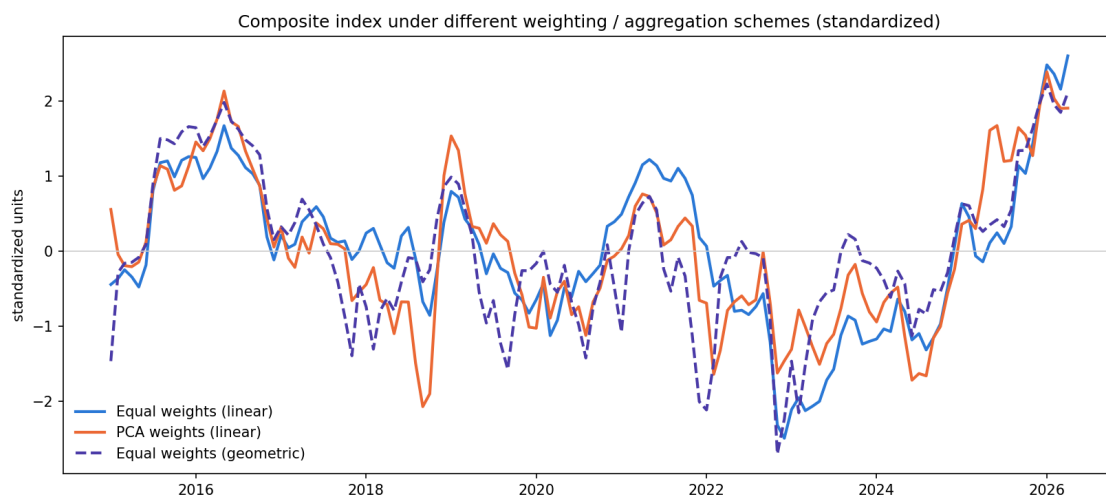


Figure 4: Composite index under equal weight, PCA weight, and geometric aggregation.